Exact Future-Archive Complexity and Exceptional-Characteristic Recovery in Typed Chronological Carry Refinement

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Abstract

We determine the exact checkpoint information needed to answer every source-dependent future query in a typed chronological carry refinement (CCR) recurrence. For freely selectable historical sources, jet degree $J \geq 8$, checkpoint $T \geq J$, horizon $H \geq 1$, and prime field $\mathbb{F}_\ell$ with $\ell > T + H$, the minimum length of an arbitrary deterministic fixed-length field-symbol archive is a direct-read count plus the rank of an explicit matrix with only five columns. The lower bound includes nonlinear encoders and is attained by a causal checkpoint representation. We also characterize when two complete histories induce the same future response profile. The boundary rank is evaluated for every admissible characteristic: horizons one to three have full rank; horizon four has exactly two exceptional prime–checkpoint pairs; horizon five is governed by a degree-six congruence; horizon six has exactly three exceptional pairs; and from horizon seven onward the boundary rank is always five. Rank losses at earlier horizons are thus restored after finitely many extra observations. The resulting archive cost eventually saturates at $5 + J(J-7)/2$, with the admissibility condition on the field understood separately at each horizon. The result extends a prior five-step CCR archive theorem without making a claim of universal machine-memory optimality or fixed-field infinite-horizon realizability.

Keywords: chronological carry refinement; exact future-response profiles; finite-field information complexity; minimal archive; exceptional primes; causal realization; polynomial recurrence.

1 Introduction

A future-query interface must preserve exactly the aspects of a past that can affect subsequent outputs when later source inputs become known. For typed chronological carry refinement (CCR), the relevant past is a source-indexed polynomial recurrence with delayed influences. A previous study [1] established the exact five-step minimum $m(T, J, 5) = 5J - 31$ under a nonvanishing-determinant hypothesis, along with a constructive archive and several distinct computational applications. The present work asks the different, general question: *what is the exact minimum archive for any prescribed horizon, including exceptional field characteristics?*

1.1 Contributions

The results are as follows.

1. A self-contained decomposition of all future response profiles into individually readable historic jet coordinates and a mandatory-only transfer quotient of dimension at most five (Theorem 4.1). It yields an exact equivalence test for complete histories, and a lower bound against nonlinear fixed-length encoders.

2. A complete characteristic-dependent rank classification for every horizon (Theorem 5.1), including two exceptional pairs at horizon four, a degree-six criterion at horizon five, and three exceptional pairs at horizon six.
3. A seventh-horizon recovery result and an explicit eventual saturation law for the archive size (Corollary 5.2). The term *saturation* is relative to each admissible horizon/field pair; one prime cannot satisfy $\ell > T + H$ for arbitrarily large H.
4. Exact determinant and resultant certificates for the exceptional-characteristic classifications, together with a paper-contained exact-arithmetic verification procedure. No external code, data file, or supplementary document is needed to reconstruct the finite checks.

1.2 Relationship to realization theory and earlier CCR work

Field-valued Hankel-rank and minimal weighted-automaton principles are classical; see [3]. Minimality and finite Hankel rank for affine LPV systems have a well-developed theory [4]. Sontag's polynomial input/output realization results [5] provide a related algebraic perspective over infinite fields. None of those general rank principles is claimed as new here. We instead evaluate the *specific time-dependent delayed CCR source-response map* exactly and classify its finite-characteristic rank defects. This paper concerns minimum checkpoint *message symbols*, not smallest internal implementation state, transient working RAM, or end-to-end execution time. In particular, the historical sources here are algebraically free, not constrained event-equality masks. The earlier five-step theorem and the CCR recurrence originate in [1, 2] and are used with attribution rather than presented as new discoveries.

2 The recurrence and archive contract

Fix integers

$$J \geq 8, \qquad T \geq J, \qquad H \geq 1, \qquad \ell > T + H \quad \text{with } \ell \text{ prime},$$

and work over $K = \mathbb{F}_\ell$ and $K[u]/(u^{J+1})$. Write $P_n = 0$ for $n < 0$, $P_0 = 1$, and for $n \geq 1$ impose the typed $p = 3$ recurrence

$$\begin{aligned} P_n = {} & (1 + (4n-3)u)P_{n-1} + 3(n-1)(4n-7)u^2 P_{n-2} - 3(n-1)(n-2)u^3 P_{n-3} \\ & + \sum_{r=4}^{\min(J,n)} \mathrm{w}_{r,n} a_{r,n} u^r P_{n-r} \pmod{u^{J+1}}, \\ & \mathrm{w}_{r,n} = \frac{(n-1)!}{(n-r)!} = \prod_{i=1}^{r-1} (n-i). \end{aligned} \tag{1}$$

Set $q_d(n) = [u^d]P_n$. For times $n \leq T$ the optional inputs $a_{r,n}$ are independently selectable in K; a history is their complete assignment. At checkpoint T, an encoder that has observed the historical source stream and its derived jets writes *exactly b symbols of K*. Only after this checkpoint is a freely chosen future source program $(a_{r,T+s})_{1 \leq s \leq H}$ revealed. A deterministic decoder, given the archive and this program, must return

$$(q_J(T+1), \ldots, q_J(T+H))$$

for every history and every future program. Encoder and decoder may otherwise be nonlinear; their transient internal computation is not charged. Let $m(T, J, H; \ell)$ be the smallest possible integer b.

The first four low-jet coefficients $q_0(n), \ldots, q_3(n)$ depend only on the mandatory part of (1); hence they are fixed, known values, independent of the optional sources.

3 Historical accessibility and direct-read information

Write

$$\mathcal{J} = \{(t,d) : 1 \le t \le T,\ 4 \le d \le \min(J,t)\}.$$

For each $(t,d) \in \mathcal{J}$ the value $q_d(t)$ is a historical high-jet coordinate.

Lemma 3.1 (Triangular accessibility)**.** *The map from freely chosen historical source amplitudes $(a_{d,t})_{(t,d)\in\mathcal{J}}$ to the tuple $(q_d(t))_{(t,d)\in\mathcal{J}}$ is a bijection $K^{\mathcal{J}} \to K^{\mathcal{J}}$.*

Proof. Proceed first by increasing time t, and at fixed t by increasing degree d. In $q_d(t)$ the *current* source $a_{d,t}$ occurs only in the summand with $r = d$, with multiplier $\mathrm{w}_{d,t}q_0(t-d) = \mathrm{w}_{d,t}$. Every other current-time source has rank smaller than d; all mandatory terms and earlier-time source contributions have already been determined. Thus

$$q_d(t) = c_{t,d} + \mathrm{w}_{d,t}a_{d,t},$$

where $c_{t,d}$ is determined by previously fixed source amplitudes. Every factor in $\mathrm{w}_{d,t}$ is a positive integer at most $T-1$, hence is nonzero in K because $\ell > T+H$. Choosing $a_{d,t} = \mathrm{w}_{d,t}^{-1}(q_d(t) - c_{t,d})$ realizes any target high-jet tuple uniquely. □

For $1 \le s \le H$ define the *direct-read* historical coordinates

$$X_{d,s} := q_d(T - J + s + d), \qquad 4 \le d \le \min(J-4, J-s). \tag{2}$$

They are historical because the defining restriction gives $T - J + s + d \le T$, and they are mutually distinct since the pair (d,s) is recoverable from the time and degree. Define

$$x(J,H) := \sum_{s=1}^{H} \max\bigl(0, \min(J-4, J-s) - 3\bigr). \tag{3}$$

Terms for $s > J-4$ vanish.

Lemma 3.2 (Complete future-dependent coordinate span)**.** *For the family of outputs $q_J(T+h)$, $1 \le h \le H$, all history-dependent coefficients of nonconstant monomials in the future source amplitudes belong to the span of the coordinate functionals $X_{d,s}$. Conversely, every $X_{d,s}$ can be recovered from the response profile of a suitable one-pulse future program.*

Proof. Use the usual weighted-path expansion of (1). Its mandatory edges have (time length, polynomial degree) $(1,0)$, $(1,1)$, $(2,2)$, $(3,3)$, and an optional source edge of rank r has (r,r). Thus each edge has time length at least its degree. On any path contributing to a future output and containing an optional future source edge, cut at the *first edge that crosses time T*. Write its old endpoint as (t,d), with $t \le T$. Because the path uses a future source edge of degree at least four, $d \le J-4$. The suffix from (t,d) to $(T+h, J)$ obeys $(T+h) - t \ge J - d$, giving

$$s := t - (T-J) - d \le h.$$

It remains to see $s \ge 1$. If the crossing edge itself is optional, with rank r and future endpoint $T+v$, the residual degree after that edge is $\delta = J - d - r \ge 0$; the predecessor has $t = T + v - r$ and hence $s = v + \delta \ge 1$. If the crossing edge is mandatory, then its predecessor has $t \ge T-2$, while $J - d \ge 4$, so $s \ge 2$. This puts the historical high-jet predecessor in (2); contributions from degrees below four are known constants. Summing paths proves the inclusion.

For the reverse inclusion, fix one index (d, s) in (2) and turn on only the optional future source $(r, n) = (J-d, T+s)$. In the output at the same time $T+s$ the response minus the all-zero-source response is precisely

$$\mathrm{w}_{J-d,T+s}\, q_d(T + s - (J - d)) = \mathrm{w}_{J-d,T+s} X_{d,s}.$$

This multiplier is a unit in K. Hence every $X_{d,s}$ is separately recoverable. Finally, outputs are multilinear in each future optional source variable along any chronological path. Evaluations on binary source assignments in $\{0, 1\}$ recover all formal monomial coefficients by inclusion–exclusion, which is valid in every field. This identifies the full span of history-dependent nonconstant response coefficients. □

4 The five-dimensional mandatory boundary

Let Z_h denote $q_J(T + h)$ evaluated with *all* future optional amplitudes zero. A first-crossing argument without future optional pulses shows that, modulo $\mathrm{span}(X)$, these mandatory-only outputs depend on just the following seven end-history cells:

$$q_J(T),\ q_{J-1}(T),\ q_{J-2}(T),\ q_{J-3}(T),$$
$$q_{J-2}(T-1),\ q_{J-3}(T-1),\ q_{J-3}(T-2).$$

The cells $q_J(T)$, $q_{J-2}(T-1)$, $q_{J-3}(T-2)$ have mandatory output coefficients independent of h, respectively 1, $3T(4T-3)$, and $-3T(T-1)$: after crossing T, they can only be followed by zero-degree one-step edges. Put

$$\begin{aligned} U_0 &= q_J(T) + 3T(4T-3)q_{J-2}(T-1) - 3T(T-1)q_{J-3}(T-2), \\ U_1 &= q_{J-1}(T), \quad U_2 = q_{J-2}(T), \quad U_3 = q_{J-3}(T), \\ U_4 &= q_{J-3}(T-1). \end{aligned} \tag{4}$$

These five coordinates are independent of X, by Lemma 3.1 and because the coefficient of $q_J(T)$ in U_0 equals one.

Define $B_H(T)$ to be the $H \times 5$ matrix of coefficients of the five original unit boundary seeds

$$q_J(T), \quad q_{J-1}(T), \quad q_{J-2}(T), \quad q_{J-3}(T), \quad q_{J-3}(T-1) \tag{5}$$

in the outputs $Z_1, \ldots, Z_H$, with all other formal historical high jets put to zero. The recurrence used in this calculation is the mandatory part of (1); its entries depend on T but not on J (for $J \geq 8$). The first column is identically one. The two discarded seed columns are proportional to it as indicated above. Hence

$$(Z_1, \ldots, Z_H)^{\mathsf{t}} = \text{known constant} + C_H X + B_H(T)U, \tag{6}$$

for an explicitly determined matrix C_H; all components are linear in the old high jets.

Theorem 4.1 (Response equivalence and optimal archive)**.** *Under the hypotheses of Section 2, two historical high-jet assignments are indistinguishable by* every *future source program through horizon H if and only if their $X_{d,s}$ values agree and*

$$B_H(T)(U - U') = 0.$$

Consequently the exact minimum fixed-length field-symbol archive is

$$\boxed{m(T, J, H; \ell) = x(J, H) + \mathrm{rank}_{\mathbb{F}_\ell}\, B_H(T).} \tag{7}$$

This lower bound holds against all deterministic encoders, linear or nonlinear; equality is attained by a causal checkpoint archive.

Proof. The nonconstant future-source coefficients of every output have, by Lemma 3.2, precisely the history-dependent span generated by the independently accessible X coordinates. Equality of all future-program responses therefore forces equality of X. With X fixed, equation (6) shows that equality for the all-zero future program is exactly the condition $B_H(U-U')=0$. Conversely these two conditions force equality of the zero-source outputs and every nonconstant future-source coefficient, hence of all program evaluations.

By Lemma 3.1 the $x(J,H)$ direct-read cells and five boundary coordinates U range freely and independently. Profile equivalence therefore has exactly $\ell^{x(J,H)+\operatorname{rank} B_H(T)}$ classes. A deterministic message of b field symbols has no more than ℓ^b values, so it must have $b \geq x(J,H)+\operatorname{rank} B_H(T)$ even for a nonlinear encoder. Retaining the X values and any $\operatorname{rank} B_H$ independent linear forms of $B_H U$ gives a matching message, from which every future response can be reconstructed by the known recurrence and its coefficient map. The archive uses only information already available at checkpoint T and may be updated causally as the historical source stream arrives. Its minimality concerns stored field symbols, not temporary execution memory. □

Remark 4.2. *The matrix B_H can be generated without enumerating historical source assignments: initialize the five seed coefficients from* (5)*, use the mandatory recurrence for $n = T+1,\dots,T+H$, and read $q_J(T+h)$. Thus* (7) *is an explicit finite computation even in exceptional characteristics.*

5 Complete characteristic-dependent rank classification

Define

$$\begin{aligned} F(T) &= 64T^6 + 1312T^5 + 11120T^4 + 49565T^3 \\ &\quad + 121776T^2 + 155655T + 80706, \end{aligned} \tag{8}$$

$$\begin{aligned} G(T) &= 256T^6 + 5632T^5 + 51040T^4 + 242500T^3 \\ &\quad + 633339T^2 + 858576T + 471216. \end{aligned} \tag{9}$$

Define finite sets of ordered pairs (ℓ, T):

$$E_4 = \{(73,13),(701,70)\}, \qquad E_6 = \{(31,23),(199,54),(3594541,1230238)\}. \tag{10}$$

The conditions $T \geq J \geq 8$ and $\ell > T+H$ are understood throughout the following theorem.

Theorem 5.1 (All admissible horizons and exceptional characteristics)**.** *Write $b_H(T;\ell) = \operatorname{rank}_{\mathbb{F}_\ell} B_H(T)$. Then*

$$b_H(T;\ell) = \begin{cases} H, & H = 1,2,3, \\ 3, & H = 4 \text{ and } (\ell,T) \in E_4, \\ 4, & H = 4 \text{ and } (\ell,T) \notin E_4, \\ 4, & H = 5 \text{ and } F(T) \equiv 0 \pmod{\ell}, \\ 5, & H = 5 \text{ and } F(T) \not\equiv 0 \pmod{\ell}, \\ 4, & H = 6 \text{ and } (\ell,T) \in E_6, \\ 5, & H = 6 \text{ and } (\ell,T) \notin E_6, \\ 5, & H \geq 7. \end{cases} \tag{11}$$

In particular, every exceptional boundary-rank deficiency disappears by horizon seven.

Proof. All minors below are determinants of the formal-seed matrix defined before Theorem 4.1. The displayed polynomials, factorizations and modular witnesses constitute the finite algebraic

certificates. Section 7 specifies how to reconstruct them directly from the recurrence without any separately supplied files.

For $H = 1$ the first row has first entry one. For $H = 2$, two 2×2 minors are $4T + 5$ and $(4T + 1)(7T + 8)$, whose resultant as polynomials in T is, up to sign, $48 = 2^4 \cdot 3$. Since $\ell > T + 2 \geq 10$, the two minors cannot vanish together. For $H = 3$, two 3×3 minors are

$$4(16T^3 + 88T^2 + 157T + 87), \qquad 3T(T + 1)(4T + 9).$$

The resultant of the nonunit factors of interest has absolute value $192 = 2^6 \cdot 3$, and $T(T + 1)$ is nonzero modulo ℓ. Hence rank three is unavoidable.

For $H = 4$, two 4×4 minors are $4f_4(T)$ and $-12T(T + 1)g_4(T)$, where

$$\begin{aligned} f_4(T) &= 1024T^6 + 14896T^5 + 88776T^4 + 272351T^3 \\ &\quad + 444443T^2 + 360948T + 114144, \\ g_4(T) &= 16T^3 + 136T^2 + 381T + 348. \end{aligned}$$

Their resultant is

$$|\operatorname{Res}(f_4, g_4)| = 2^{29}3^3 \cdot 29 \cdot 73 \cdot 701.$$

The only joint modular roots at these candidate primes are $T \equiv 0 \pmod{29}$, $T \equiv 13 \pmod{73}$, and $T \equiv 70 \pmod{701}$. Because $8 \leq T < \ell$, the first is impossible. For the remaining two pairs, all five maximal 4×4 minors vanish; rank is exactly three by the $H = 3$ result. This proves the horizon-four classification.

For $H = 5$, the square determinant is the previously identified CCR mandatory-transfer determinant [1]:

$$\det B_5(T) = 192T(T + 1)F(T).$$

Thus its vanishing is equivalent to $F(T) \equiv 0 \pmod{\ell}$. Whenever the first four rows have rank four, B_5 has rank at least four. At the two exceptional horizon-four pairs, the 4×4 minor with rows $1, 2, 3, 5$ and columns $1, 2, 3, 4$ has residues 12 modulo 73 and 575 modulo 701, respectively. Therefore rank cannot be smaller than four, proving the exact horizon-five criterion.

For $H = 6$, two of the six maximal 5×5 row-minors are

$$192T(T + 1)F(T), \qquad 192T(T + 1)G(T).$$

An independently calculated integer Sylvester resultant gives

$$|\operatorname{Res}(F, G)| = 2^{26}3^4 5^5 \cdot 31 \cdot 73 \cdot 199 \cdot 701 \cdot 3594541.$$

The admissible characteristic eliminates $2, 3, 5$. Modular polynomial gcd calculations yield exactly the five candidate pairs

$$(31, 23), \quad (73, 13), \quad (199, 54), \quad (701, 70), \quad (3594541, 1230238).$$

Checking the *remaining* maximal row-minors excludes $(73, 13)$ and $(701, 70)$; all six vanish for the three pairs in E_6. A 4×4 leading minor is nonzero at each remaining pair, with residues 17, 20 and 1907807 in the same order, so the rank is exactly four there. The appended seventh row restores rank five: the 5×5 determinant using rows $1, 2, 3, 4, 7$ and all five columns has residues 26, 171, and 1970365 modulo 31, 199, and 3594541, respectively. These are nonzero exact certificates. At every other admissible parameter, the first six rows already have rank five. Therefore $H \geq 7$ always has boundary rank five, completing the proof. □

Corollary 5.2 (Exact saturation and recovery)**.** *For every admissible parameter set, the exact archive size is obtained by inserting* (11) *into* (7)*. In particular, for* $H \geq 7$*,*

$$m(T, J, H; \ell) = 5 + \sum_{s=1}^{H} \max\bigl(0, \min(J-4, J-s) - 3\bigr). \tag{12}$$

For $H \geq \max(7, J-4)$*, this simplifies to*

$$\boxed{m(T, J, H; \ell) = 5 + \frac{J(J-7)}{2}.} \tag{13}$$

Proof. Theorem 5.1 gives $b_H = 5$ for $H \geq 7$. In (3), for every $d = 4, \ldots, J-4$ the allowable values of s are $1, \ldots, J-d$. The sum of their counts is $\sum_{d=4}^{J-4}(J-d) = J(J-7)/2$; when $J = 8$ the same equality holds with one term. Thus the direct-read contribution ceases growing by horizon $J-4$. □

Remark 5.3 (No fixed-field infinite-horizon assertion)**.** *The statements hold for each instance with* $\ell > T + H$*. If* T, J, ℓ *are fixed, admissible horizons have* $H < \ell - T$*. The phrase* eventual saturation *means that once the threshold lies within this admissible range, extending the horizon while preserving the stated assumptions requires no further field symbols. It does not assert one fixed field can support arbitrarily large* H *under the same accessibility proof.*

6 Examples and information interpretation

For $J = 12$ and generic admissible characteristics, formula (7) gives

Horizon H	4	5	6	7	8 and higher
Archive symbols	24	29	32	34	35

Here $H = 5$ reproduces the prior five-step theorem; the new result explains the subsequent growth and the exact plateau. For the exceptional case $(T, J, \ell) = (23, 8, 31)$, admissible horizons $5, 6, 7$ have costs $(8, 8, 9)$ rather than the corresponding generic values $(9, 9, 9)$. Thus an exceptional characteristic can temporarily remove a response direction, and an additional future observation can restore it.

The response-equivalence description is more informative than a mere number of symbols. It says exactly which differences in an old history can never be resolved by future programs through horizon H: they must vanish on all $X_{d,s}$ and lie in the kernel of the mandatory transfer. This finite characteristic kernel may depend on (ℓ, T, H); the direct-read coordinate family does not.

7 Reconstruction of the algebraic certificates

All algebraic inputs required for an independent check are stated in this article. In particular, no executable code or separate certificate file is needed. The following exact-arithmetic procedure reconstructs the finite computations used in Theorem 5.1.

1. Work over $\mathbb{Z}[T]$ and initialize five formal boundary seeds corresponding, in order, to the cells in (5), with all other historical high-jet seeds equal to zero. Use the mandatory three-lag recurrence in (1) to propagate the five response columns. Extract the coefficient of degree J at times $T+1, \ldots, T+7$. Because only offsets from the terminal degree occur, the resulting rows of $B_7(T)$ do not depend on $J \geq 8$.

2. Compute the maximal minors of $B_4(T)$ and $B_6(T)$ by exact determinant expansion, fraction-free elimination, or another exact method. Check the two displayed fourth-horizon minor factors $4f_4(T)$ and $-12T(T+1)g_4(T)$ and the first two sixth-horizon factors $192T(T+1)F(T)$ and $192T(T+1)G(T)$. All the defining polynomials and their coefficients appear in Section 5.
3. Form the Sylvester matrices for (f_4, g_4) and (F, G) over $\mathbb{Z}$. Their determinants have the prime factorizations shown in the proof. For each candidate prime, reduce both polynomials in $\mathbb{F}_\ell[T]$ and compute their polynomial gcd. The common roots are explicitly enumerated in the proof, so no numerical root approximation is necessary.
4. At every surviving prime–checkpoint pair, reduce the entire transfer matrix modulo ℓ and use exact row reduction to test all maximal minors. The stated nonzero lower-order minor residues certify the precise rank drops. For each of the three sixth-horizon exceptions, the nonzero determinant residues 26, 171, and 1970365 for rows $1, 2, 3, 4, 7$ certify the seventh-horizon rank recovery.
5. Finally, obtain any archive size by computing the finite integer sum (3) and adding the exact modular rank specified by (11). The examples of Section 5 and the final section can be checked by this rule without reconstructing source-history data or using approximate arithmetic.

The all-horizon archive theorem is proved by the accessibility, path decomposition, and profile-counting arguments of Sections 3–4; finite determinant checks are used only to evaluate the exceptional-characteristic rank profile. No external datasets were used.

8 Discussion

The exact rank profile shows that CCR's source chronology can force a sharp separation between individually interrogable historical coordinates and the small mandatory transfer left after quotienting them out. The resulting five-column matrix makes exceptional field behavior explicit and supplies an optimal checkpoint message for the stated free-source model. These results extend the earlier five-step theory, but they do not establish optimal peak RAM, universal algorithmic speedups, optimal archives for event-derived constrained sources, or any statement about composite moduli. Extending accessibility beyond the characteristic constraint is a separate question and cannot be assumed without proof.

Statements and Declarations

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Data availability: No external datasets were used. All definitions, parameters, polynomials, finite certificate values, and required verification steps are included in this article. No supplementary dataset is needed.
Code availability: No source code, executable program, or separate proof supplement accompanies this article. The exact algebraic calculations can be reconstructed independently from the stated recurrence, determinant identities, modular witnesses, and the verification procedure in Section 7. No unpublished program is required for the mathematical arguments.

References

[1] A. E. Bütün, *The Computational Power of Typed Chronological Carry Refinement: Sharp Forecast-Information Bounds, Fast Exact Algorithms, Finite-Ring Profiles, and Lossless Source Coding*, manuscript (2026).

[2] A. E. Bütün, *CCR-k General Theory: Higher-Order Chronological Carry, Hessenberg Hierarchies, and Universal Stirling Shadows*, preprint (2026), DOI: 10.5281/zenodo.22734188.

[3] S. Kiefer, *Notes on Equivalence and Minimization of Weighted Automata*, arXiv:2009.01217 (2020), `https://arxiv.org/abs/2009.01217`.

[4] M. Petreczky, R. Tóth, and G. Mercère, "Realization theory for LPV state-space representations with affine dependence," *IEEE Trans. Automat. Control* **62**(9), 4667–4674 (2017), DOI: 10.1109/TAC.2016.2629989.

[5] E. D. Sontag, *Discrete-time polynomial systems with inputs and outputs: structure, reachability, observability, and minimal realizations*, arXiv:2609.36423 (2026), `https://arxiv.org/abs/2609.36423`.